Hsuan-Chi Liu

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Education

National Taiwan University Taipei, Taiwan

B.S. in Computer Science and Information Engineering; GPA: 3.94/4.00 (4.15/4.30) *Sep 2022 – Jun 2026*

- **Research Interests:** Efficient Machine Learning, High-Performance Computing, Generative Modeling

Publications

[*] Equal contribution; [†] Corresponding author.

Learning Conditional Source Distribution via Flow Reversal for Temporal Flow Matching
Kuan-Hsun Tu[*], **Hsuan-Chi Liu**[*†], Jia-Wei Liao[*], Chiensheng Chiang, Cheng-Fu Chou, and Tsung-Wei Ke.
In submission, 2026.

TeCoP: Accelerating Diffusion Large Language Model Decoding via Tentative Context Pipelining
Jui-Chien Tsou[*], **Hsuan-Chi Liu**[*], Wei-Chin Wang[*], Yu-Syuan Xu, and Chun-Yi Lee.
In submission, 2026.

Boosting Small Object Tracking via Collaborative Detection Transformer
Bing-Cheng Chuang[*], Jia-Hua Lee[*], Pin-Tseng Kuo[*], I-Hsuan Chu[*], Yi-Shein Hsiao[*], Cheng-Han Wu[*], Po-Yi Wu[*], Jui-Chien Tsou[*], **Hsuan-Chi Liu**[*], Chun-Yi Lee, and Yuan-Fu Yang.
19th International Conference on Machine Vision and Applications (MVA), IEEE, 2025.

Research Experience

Embodied AI Lab, National Taiwan University

Advisor: Prof. Tsung-Wei Ke *Nov 2024 – Present*

- **Flow Matching for Video Prediction.** Developed *CNP-Flow* to learn condition-dependent source distributions through flow reversal, improving both generation quality and efficiency.
- **VLM-guided exploration in Robot Reinforcement Learning.** Designed a collaborative VLM–VLA framework to guide action sampling and improve the sample efficiency of RL exploration.

ELSA Lab, National Taiwan University

Advisor: Prof. Chun-Yi Lee *Oct 2024 – 2026*

- **Diffusion Language Model Decoding.** Leveraged intermediate predictions as additional contextual information to enable faster decoding across blocks.
- **Parallel Image-Stitching Optimization.** Accelerated SIFT preprocessing with OpenMP and offloaded Gaussian blur to GPUs using ROCm HIP.

MIT Archi Team & CERN CMS

Research Collaboration *Jul 2026 – Oct 2026*

- **Knowledge Retrieval for LLM Support Agents.** Evaluated operational knowledge-graph retrieval agent against existing baselines using 178 support tickets and 69 FAQs with gold evidence annotations.

Institute of Information Science, Academia Sinica

Advisor: Prof. Meng-Tsung Tsai *Jun 2023 – Aug 2023*

- **Minimum Cycle Cover in Streaming Graphs.** Explored approximation algorithms to reduce the space and time required for graph processing.

Awards & Competitions

Overall Champion, Student Cluster Competition (SCC25) Nov 2025

ACM/IEEE Supercomputing Conference *St. Louis, USA*

- Ran MLPerf Inference (Llama 3.1 8B) on 2-node H200 GPUs via Triton, achieving highest throughput among all teams under a 10 kW power cap.
- Identified performance bottlenecks in the Exascale Climate Emulator through profiling, including redundant data-type conversions and GPU underutilization.

Second Place & Best AI Performance Award Nov 2025

APAC HPC-AI Competition, Asia Pacific Advanced Network *Singapore*

- Optimized MPI parallelization for the NWChem benchmark to reduce runtime and improve scalability.

Third Place, SMOT4SB Challenge (MVA2025) May 2025

19th International Conference on Machine Vision and Applications (MVA) *Kyoto, Japan*

First Place & Group Competition Award May 2025

ASC Student Supercomputer Challenge, Asia Supercomputer Community *Qinghai, China*

Work Experience

CERN Summer Student (CMS Offline & Computing) Jun 2026 – Aug 2026

European Organization for Nuclear Research (CERN) *Geneva, Switzerland*

- Developed MCP-based diagnostic tools to ground LLM agents in real-time CMS computing data, enabling evidence-based reasoning and automated troubleshooting.

Wireless Team Lead Aug 2025 – Jun 2026

Network Administration Team, NTU CSIE *Taipei, Taiwan*

- Administered the department-wide wireless network and collected logs for monitoring and troubleshooting.

Teaching Assistant (Parallel Programming) Sep 2025 – Dec 2025

National Taiwan University *Taipei, Taiwan*

Summer Research Intern Jun 2023 – Aug 2023

Institute of Information Science, Academia Sinica *Taipei, Taiwan*

Open-Source Contributions

MLCommons – MLPerf Inference 5.0 Nov 2025

- Contributed multi-node LLM inference configurations and documentation using NVIDIA Triton Inference Server for distributed MLPerf benchmarking.

Skills

Languages: Python, C/C++

GPU Acceleration: CUDA, cuBLAS/cuBLASLt, ROCm HIP

Parallel Computing: MPI, OpenMP

Profiling & Benchmarking: Nsight Systems, MLPerf Inference, multi-node scaling & tuning

Systems & Tools: Linux, Git, Docker, Slurm, OPNsense, DNS, LLM Agents, MCP, OpenSearch